

SDI Intelligence server: who does what

Jack builds the machine and hands over a Windows 11 virtual machine that behaves like a new laptop on the network, with the graphics card inside it. James takes it from there, exactly as he would set up a laptop. The handover checks in the middle are the line between the two.

Scan 3XS · Threadripper PRO 9975WX (32 cores) · 128 GB ECC · NVIDIA RTX 4000 Ada SFF 20 GB · 2 × 2 TB NVMe · Windows Server 2025 host · Windows 11 VM for SOLIDWORKS

JACK Infrastructure: the host and the VM

Everything below the applications: hardware, operating systems, network, accounts, backup. In build order.

THE HOST (WINDOWS SERVER 2025)

1. **BIOS:** latest firmware; enable virtualisation (SVM) and IOMMU, which the GPU pass-through needs.
2. **Storage:** set up the two NVMe drives (mirror or split OS / VMs — Jack's call) and install Windows Server 2025 with current drivers.
3. **Network and domain:** fixed IP, DNS name, join the domain.
4. **Hyper-V:** add the Hyper-V role.
5. **Service account** for the SDI Intelligence portal service, with read/write to the estimating shares by UNC path (`\\server\share` , not K:).
6. **Firewall:** allow the portal port (`8071`) from the office network and from the VM.

THE VM (WINDOWS 11)

7. **Create it as Generation 2** with Secure Boot and a virtual TPM (Windows 11 requires both).
8. **Fixed resources:** 12 virtual CPUs, 48 GB memory with Dynamic Memory **off**, a 300 GB+ virtual disk on NVMe.
9. **Pass the RTX 4000 through to the VM** (Discrete Device Assignment), then install NVIDIA's RTX workstation driver inside the VM. GPU partitioning is not needed.
10. **Windows 11 Pro**, fixed IP and DNS name, joined to the domain, fully updated.
11. **Runner account:** a dedicated domain user that logs on automatically at boot; read/write to the estimating shares by UNC path.
12. **Power and lock policy:** no sleep, no screen lock, no screensaver on the VM — SOLIDWORKS and Excel need a live desktop session.
13. **Admin access for James** on both machines: Remote Desktop, and the host's admin share (the laptop's `push-to-server.ps1` copies through it).

RUNNING IT

14. **Backups:** host and VM, with one restore tested before go-live.
15. **Windows Update:** restarts scheduled out of hours, host and VM on different nights.
16. **Antivirus:** apply the exclusions James supplies (SOLIDWORKS temp and the output folders).
17. **Power:** on the UPS, and set to start again after a power cut, with the VM starting automatically.

Handover checks

Jack confirms these, then the machine is James's:

- In the VM's Device Manager the graphics card shows as **NVIDIA RTX 4000 Ada** with the NVIDIA driver, not "Microsoft Basic Display".
- After a full restart of the host, the VM starts and logs in as the runner account with nobody touching it.
- Signed in as the runner account, the VM can open and save a file in the estimating folders by UNC path.
- From the VM, the host answers on port 8071 (once James's service is running, this is just a browser test).
- James can Remote Desktop into both, as an administrator.

JAMES

Applications: the laptop premise

Everything that makes it an SDI Intelligence machine, installed the same way as on the laptop.

ON THE VM (RUNS THE ESTIMATES)

1. **SOLIDWORKS:** install and activate (see licensing below), open one model to confirm the graphics card is in use.
2. **Office / Excel:** install and activate; open one estimate book.
3. **PDM / Design Vault client,** if models come from the vault.
4. **Python and Git,** clone the repository, create the engine's `.venv`.
5. **The `.env` file,** written by hand for this machine — never copied from the laptop.
6. **The runner:** scheduled task at logon (`restart-runner.ps1`) pointing at the host's service.
7. **Hand Jack the antivirus exclusion paths.**

ON THE HOST (RUNS THE PORTAL)

8. **Python** and the service's own `.venv`, the backend files, and the host's own `.env`.
9. **Install and start the SDIIntelligence service;** check the portal loads from a desk.

PROVING IT

10. **Run one known job** (12645 or 12173) on the VM and compare the book and the run time with the laptop's.

11. **Restart the host** and confirm the portal and the runner both come back without anyone logging in.

Agree together before the build week

Rule of thumb: if it would be the same on any Windows machine, it's Jack's. If it only exists because this is SDI Intelligence, it's James's.

QUESTION	WHO LEADS	WHY IT MATTERS
SOLIDWORKS licence type: standalone or network (SolidNetWork), and cover for unattended automated use	James, with the reseller	A standalone licence is tied to the machine and can need reactivating when a VM changes; a network licence avoids that. If it's a network licence, Jack hosts the licence server.
SOLIDWORKS support for this setup: your version on Windows 11 24H2, in a Hyper-V VM with the card passed through	James, with the reseller	Unsupported setups often run fine, but support can decline problems on them.
Machine names and IP addresses for host and VM	Jack	The runner, the portal links and the service settings all name them.
Who is told before Windows updates or restarts	Jack	A restart mid-run loses that estimate.